

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- | | | |
|---|-----|---|
| A) State and explain any six features of python. | CO1 | 6 |
| B) Explain input and print command in python with suitable example. | CO1 | 6 |
| C) What will be the output of following print statements? | CO2 | 6 |
| I. <code>print('GoodLuck' * 2)</code> | | |
| II. <code>print(3-4j + 2+6j)</code> | | |
| III. <code>print('3+4j'*int(4/2))</code> | | |

Q.2 Solve Any Two of the following.

- | | | |
|--|-----|---|
| A) Explain following operators of python with suitable examples of each. | CO1 | 6 |
| I. is | | |
| II. in | | |
| III. not in | | |
| B) Explain while loop with suitable example? | CO1 | 6 |
| C) The distance between two cities (in km.) is input through the keyboard. | CO2 | 6 |
| Write a python program to convert and print this distance in meters and centimeters. | | |

Q. 3 Solve Any Two of the following.

- | | | |
|--|-----|---|
| A) Explain break, continue and pass statement with suitable example of each. | CO1 | 6 |
| B) Explain <code>append()</code> and <code>copy()</code> methods of list with suitable examples of each. | CO1 | 6 |
| C) The length & breadth of a rectangle are input through the keyboard. | CO2 | 6 |
| Write a python program to calculate and print the area & perimeter of the rectangle. | | |

Q.4 Solve Any Two of the following.

- | | | |
|---|------------|----------|
| A) Explain “and, or, not” operators of python with suitable examples of each. | CO1 | 6 |
| B) Write a python program to print following pattern (using while loop).
aaaaaaa
aaaaa
aaa
a | CO3 | 6 |
| C) What are tuples? With suitable example explain how to add elements to a Tuple. | CO1 | 6 |

Q.5 Solve Any Two of the following.

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|--|------------|----------|
| A) What are functions in python? Explain recursion with suitable example. | CO1 | 6 |
| B) What is package? What is Python PIP? With example explain how to get list of all packages installed on your system? | CO2 | 6 |
| C) What will be the output of following python statements?
I. <code>print(23 // 5)</code>
II. <code>print(2 << 2)</code>
III. <code>print(2 >> 0)</code>
IV. <code>print(2 ^ 2)</code>
V. <code>print(2 != 2)</code>
VI. <code>print(2 < 0)</code> | CO3 | 6 |

***** End *****